# A HIGH-THROUGHPUT NANOFLUIDIC DEVICE FOR SMALL EXTRACELLULAR VESICLE NANOPORATION

*Rui Hao[1,2], Zitong Yu[1], Jing Du[1], Hang Guo[2], Yi Zhang[3,4], and Hui Yang[1,3,*]*
[1]Laboratory of Biomedical Microsystems and Nano Devices, Bionic Sensing and Intelligence Center, Institute of Biomedical and Health Engineering, Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, Shenzhen, China
[2]Pen-Tung Sah Institute of Micro-Nano Science and Technology, Xiamen University, Xiamen, China
[3]CAS Key Laboratory of Health Informatics, Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, Shenzhen, China
[4]Center for Medical AI, Institute of Biomedical and Health Engineering, Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, Shenzhen, China

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## ABSTRACT

We propose a nanofluidic device named "small extracellular vesicle nanoporator (sEVN)" for high-throughput nanoporation of small extracellular vesicles (sEVs). By transporting sEVs through constriction nanochannel with height comparable to their dimension, transient nanopores are introduced on the sEVs membrane by mechanical membrane disruption, while maintaining sEV integrity. The sEVN consisting of 30,000 nanochannels demonstrates a high sample throughput and absence on clogging issues, and the working mechanism of the device is elucidated through experimental and numerical study. This platform provides a way for high-throughput sEV nanoporation and offers promises for engineering sEVs as universal drug delivery vehicles for biomedical applications.

## KEYWORDS

Nanofluidics, High throughput, Small extracellular vesicle, Nanoporation, Membrane disruption

## INTRODUCTION

sEVs are lipid-bilayer nanoscale membrane particles (< 200 nm in size) actively secreted by numerous cell types [1]. Since sEVs exhibit a naturally derived composition and function as intercellular communication tools, they have great potential as universal drug delivery vehicles for biological research and medical applications [2]. In order to load exogenous biomolecules into sEVs, nanopores served as drug transport channels need to be introduced on the sEV membrane by engineering methods [3].

Nanofluidics can manipulate physical and biological objects of interest in nanoscale, therefore holding great potential to sEV nanoporation [4]. Due to its unique ability, nanofluidics have been widely used in manipulation of nanoparticles, translocation of DNA molecules, tracking of viruses [5]. However, to the best of our knowledge, there is still no report on sEV nanoporation by using nanofluidic devices yet. The intrinsically small feature size of the nanofluidic devices bring difficulties when high sample throughput is required, and such requirement is essential for applying engineered sEVs in biomedical studies.

In this paper, we present a novel nanofluidic device named "sEVN" for introducing transient nanopores onto sEVs by mechanical membrane disruption (Figure 1A). An array of 30,000 nanochannels with height comparable to the dimension of sEVs is used for sEV nanoporation with high sample throughput, avoiding clogging issue inside the channels. Meanwhile, a numerical study of the fluidic field points out the velocity gradient and fluidic pressure. The former correlates to the fluid shear stress that determines together with mechanical compression the sEV nanoporation mechanism. The latter is related to the PDMS-glass bonding strength that determines the allowable flow rate during the nanoporation process. Furthermore, successful experiment results demonstrate that this technique will not only boost the field of nanofluidic technology for mechanical membrane disruption of sEVs but also open new opportunities to develop sEV vehicles for biomedical applications.

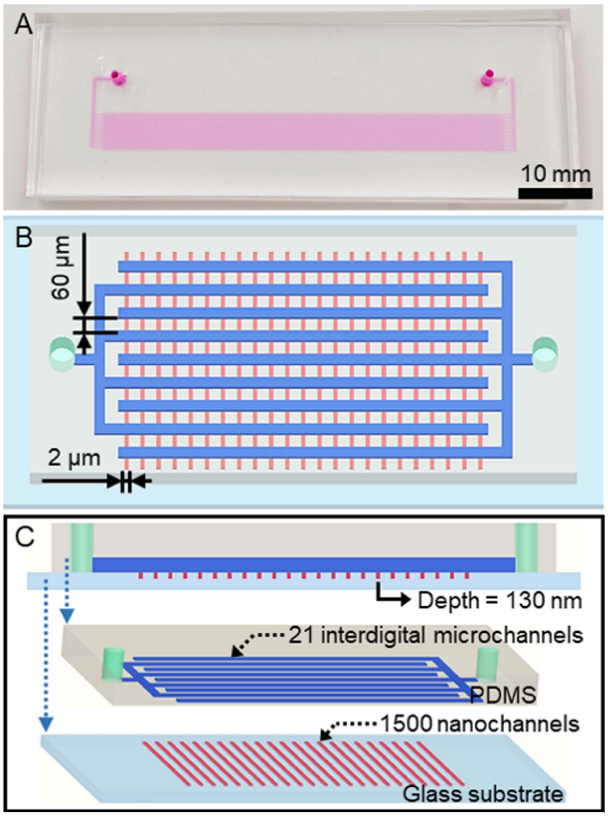


*Figure 1: Picture and schematic illustrations on the sEVN. (A) Photograph of the sEVN device. (B) Layout of the interdigital microchannels and the nanochannel array. (C) Schematic illustration of the device consisting of a PDMS layer with 21 interdigital microchannels and a glass substrate with 1500 parallel nanochannels.*



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## MATERIALS AND METHODS

### Materials

Materials used for nanofluidic device fabrication include glass substrates (Schott AG, USA), silicon wafers (RDMICRO, China), SU-8 photoresist and developer (MicroChem, USA), S1805 photoresist and developer (Rohm and Haas Electronic Materials, USA), and PDMS (Dow Corning, USA). All reagents required for cell culture are purchased from Gibco (USA), including Ham's F-12K (Kaighn's) medium (F-12K), fetal bovine serum (FBS), penicillin-streptomycin, phosphate buffered saline (PBS), etc. For sEV isolation, the relevant materials include exosome-depleted FBS (Shanghai XP Biomed, China) and Exo-Spin Exosome Purification Kit (Cell Guidance Systems, USA).

### Cell Culture and sEV Isolation

The human non-small cell lung cancer cell line of A549 is purchased from Stem Cell Bank, Chinese Academy of Sciences (China), and cultured with F-12K medium supplemented with 10% FBS and 1% penicillin-streptomycin in a cell incubator at 37 °C in 5% $CO_2$. When cell density reached 60% confluence, the cells are washed 3 times with PBS and then cultured in F-12K supplemented with 10% exosome-depleted FBS to prevent contamination of exogenous sEVs. After 48 hours, cell culture medium is collected from cell culture flask. sEVs are isolated from the medium using the Exo-Spin Exosome Purification Kit according to the operation instructions provided by the manufacturer (Cell Guidance Systems, USA).

### Nanofluidic Device Design and Fabrication

In this study, the sEVN layout is such that in- and outlet microchannels contact an array of nanochannels with well-defined dimensions (Figure 1B). In order to improve the device throughput, 21 parallel microchannel are fabricated on a PDMS layer to generate an interdigital layout with 10 microchannels connected to the inlet and 11 microchannels connected to the outlet, and the interspace between two adjacent interdigital microchannels is 60 μm. 1500 parallel nanochannels are fabricated on a glass substrate with the orientation perpendicular to that of the microchannels, and the PDMS layer is bonded to the glass substrate (Figure 1C). In our design, each pair of inlet and outlet microchannels is connected to 1500 parallel nanochannels, which is equivalent to in total 30,000 (i.e. 1500 × 20) nanochannels. Moreover, the nanochannel width is set to 2 μm to further improve the device throughput, while the optimum depth of the nanochannel array is 130 nm, which is comparable to the dimension of the sEV samples.

The sEVN fabrication is modified from previous published works [4,5]. As illustrated in Figure 2, the PDMS layer with microchannels is replicated from a SU-8 structure that is fabricated on a silicon wafer by soft lithography. Next, the nanochannel array is fabricated on the glass substrate by reactive ion etching (RIE) with standard cleanroom processes. Finally, the PDMS layer and the glass substrate are treated by oxygen plasma (PDC-MG, WEIKE, China) and bonded together to obtain the device.

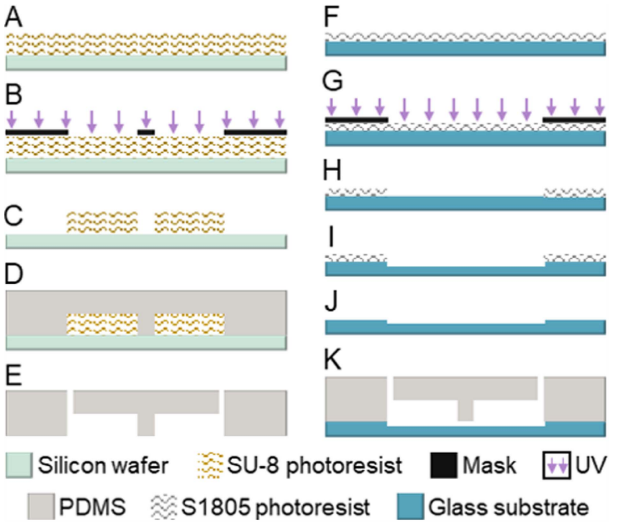


*Figure 2: Schematic illustration of the device fabrication. (A-C) SU-8 is spin-coated on the silicon wafer, patterned by UV exposure and then developed. (D-E) PDMS is casted on the SU-8 mold, peeled off and punched to make access holes. (F-H) S1805 photoresist is spin-coated on the glass substrate, patterned by UV exposure and then developed. (I) Glass substrate is etched by RIE. (J) Photoresist residue is then removed. (K) The PDMS layer and the glass substrate are bonded by oxygen plasma treatment.*

### Numerical Simulation

The numerical study of the flow field distribution and the pressure distribution in sEVN nanochannels are carried out by finite element method (FEM) in COMSOL Multiphysics (COMSOL 5.3a, COMSOL, Sweden). The Navier-Stokes equations are used to study the flow field. Considering that the width of the nanochannel is much larger than the depth, a two-dimensional model along the cross-section of the nanochannel is built to discuss the properties of the device. In this model, the non-slip boundary condition is used at all channel walls, the fluidic density and dynamic viscosity are set to those of water at 293.15 Kelvin, the flow rate boundary condition is applied at the inlet, and the null pressure condition is applied at the outlet.

### Nanoporation of sEV Membrane

After the device being ready, the isolated sEVs are dispersed in PBS and then transported into the sEVN using a syringe pump (neMESYS, Cetoni, Germany) at different flow rates (0 – 12 μL $min^{-1}$). Then, the sEVs are mechanically deformed as they rapidly pass through the nanochannels, favoring membrane disruption and introducing transient nanopores on the membrane by mechanical compression and fluid shear. Thereafter, the treated sEVs are collected at the outlet of the device for further analysis and applications.

## RESULTS AND DISCUSSIONS

### Nanochannel Topography of the Nanofluidic Device

Figure 3A shows the microscopic image of the nanochannel array contacting the adjacent microchannels, and the layout of the sEVN is equivalent to 30,000 nanochannels working simultaneously. The amount of



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nanochannels is increased ~ 300-fold in comparison with the conventional nanofluidic device [6]. The cross-section of a nanochannel is shown in Figure 3B. Correspondingly, the nanochannel width measured by scanning electron microscopy (SEM, Gemini SEM, Zeiss, Germany) and the nanochannel depth measured by atomic force microscopy (AFM, Agilent 5500 AFM, Agilent Technologies, USA) are shown in Figure 3C – 3E, respectively. The measurements show that the width of the nanochannel is 1.9 µm with the variation below ± 0.1 µm over the whole glass substrate surface and the depth of the nanochannel is 130.7 nm ± 0.6 nm. After bonding the glass substrate to the PDMS layer, the aspect ratio of the nanochannel is approximately 15 (width/depth), which is larger than that of previous published works [5,6]. The device works without PDMS collapsing onto the bottom surface of the nanochannel, this ensures high-throughput sample processing capability, and avoids clogging issues as well. Furthermore, the surface roughness of the nanochannel is shown in Figure 3F. The surface roughness on the bottom of nanochannel, which is defined as the root mean square height is below 1.8 nm, the low roughness of the surface can effectively avoid damaging the integrity of sEVs.

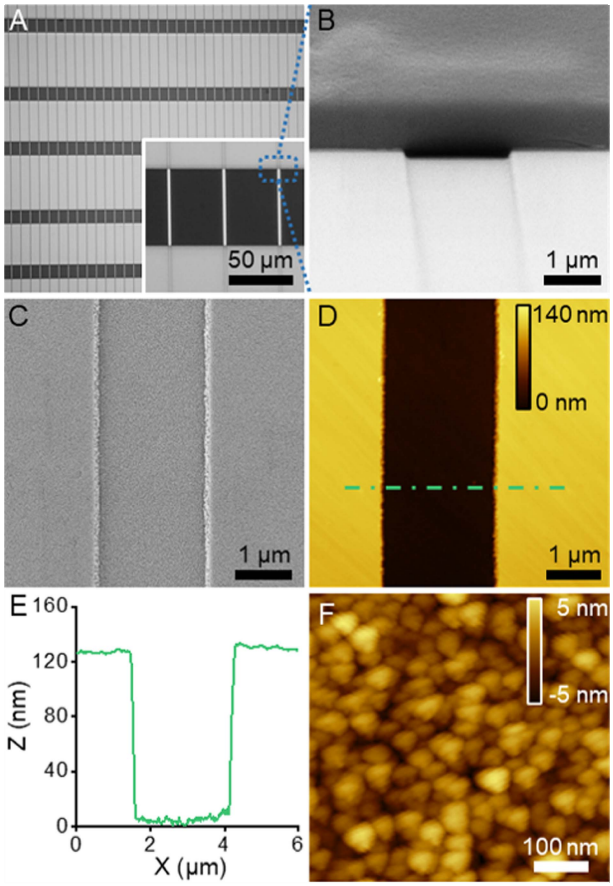


*Figure 3: Characterization of the sEVN device. (A) Microscopic image on the parallel nanochannel array with the inset showing a higher magnification image of the nanochannels contacting adjacent microchannels. (B) SEM image on the cross-section of the nanochannel. (C) SEM image showing the nanochannel width. (D) AFM nanochannel topography. (E) The profile along the green dashed line in D. (F) AFM measurement on the surface roughness of the nanochannel bottom.*

**Analysis of Nanofluidic Field Inside a Nanochannel**

The sEV transporting through the sEVN is elucidated by FEM study in the vicinity of the nanochannel at different flow rates (e.g., 8 µL min$^{-1}$), as shown in Figure 4A. During the process, the flow velocity shows a parabolic profile along the depth of the nanochannel and the maximum velocity gradient increases with the flow rate (Figure 4B and 4C). Because of the positive correlation between velocity gradient and fluid shear stress according to the Newton's law of viscosity [7], more and/or bigger nanopores can be introduced on the sEV membrane, indicating a better nanoporation efficiency at a higher sample flow rate. However, when the flow rate is increased to 12 µL min$^{-1}$, the FEM results reveal that the pressure acting on the interface between the PDMS layer and the glass substrate exceeds the maximum PDMS-glass bonding strength of 850 kPa [8], easily causing the sample leakage in the device. Therefore, the 8 µL min$^{-1}$ is chosen as the optimal flow rate for sEV nanoporation in the upcoming biological studies.

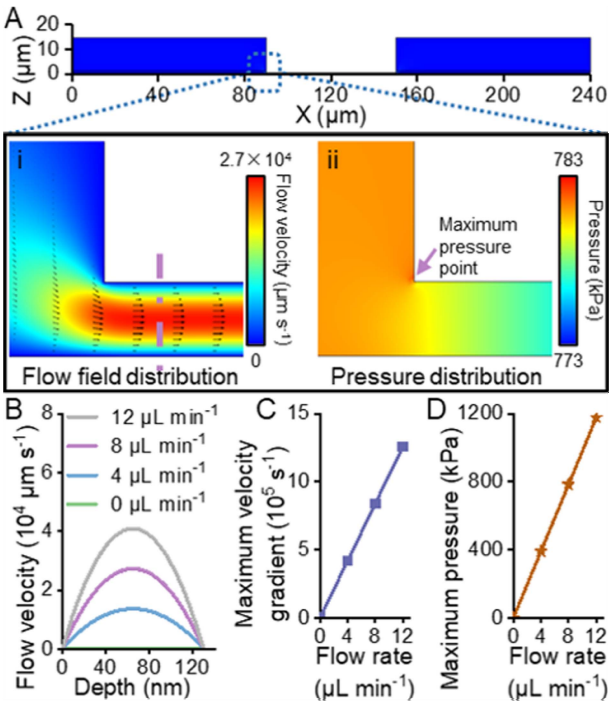


*Figure 4: FEM simulation of the flow field in the nanofluidic device. (A) Illustrations on the FEM model, showing (i) flow field distribution and (ii) pressure distribution in the sEVN device at 8 µL min$^{-1}$. (B) Flow velocity profiles at different flow rates along the purple dashed lines in (A). (C) Corresponding maximum velocity gradient of (B). (D) Plot of maximum pressure at different flow rates.*

**Assessment of sEV Nanoporation**

In order to evaluate the nanoporation, sEVs before and after being treated by the sEVN are characterized using a nanoparticle tracking analysis (NTA) instrument (ZetaView PMX 110, Particle Metrix, Germany). During the nanoporation procedure, the sEVs are introduced into the device at flow rate of 8 µL min$^{-1}$ with a total volume of 100 µL, which is a necessary sample volume for NTA. As a result, the untreated sEVs show a narrow size distribution



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from 100 nm – 200 nm (Figure 5A and 5B), with 65.1% sEVs bigger than 130 nm (Figure 5C). Moreover, the size of the sEVN-treated sEVs shows an increasing trend, the maximum concentration appears at the sEV size of 170 nm instead of 130 nm (Figure 5B), which is consistent with previously report nanoporation methods [2]. The proportion of sEVs bigger than 130 nm increases to 82.5% (Figure 5C), also representing the nanoporation effect of the sEVs treated by the sEVN device [9]. The increase on the sEV size distribution after transporting the sEVs through the device indicate the phenomenon of loading exogenous materials to sEVs. In addition, the results show that the sEV concentrations before and after being treated by the sEVN are $6.4 \times 10^9$ sEVs $mL^{-1}$ and $6.1 \times 10^9$ sEVs $mL^{-1}$, respectively, corresponding to a recovery yield of 95.3%. Thus, the clogging issues of large-size sEVs inside the nanochannels can be ignored. Notably, when the flow rate of 8 μL $min^{-1}$ is used in this study, there is no cracks and sample leakage in the device. Comparing to the conventional nanofluidic device, the flow rate used in the nanoporation procedure is increased ~ $10^4$-fold [10].

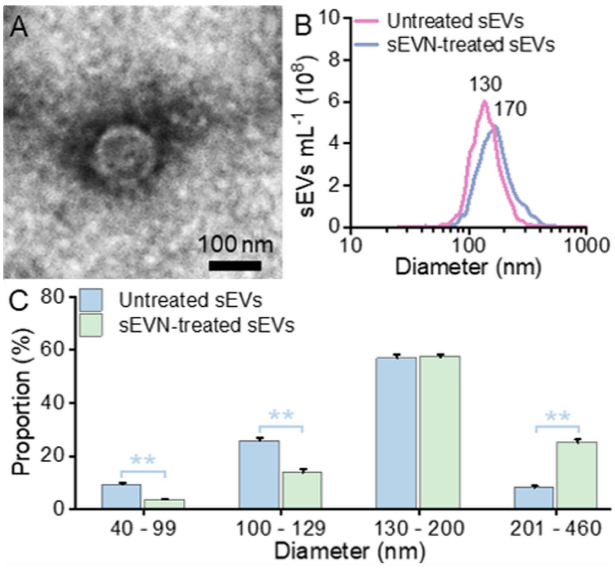


*Figure 5: Characterization of untreated and sEVN-treated sEVs. (A) Transmission electron microscopy image of untreated sEVs. (B) Size distribution of untreated and sEVN-treated sEVs obtained by NTA. (C) Histogram on the size of untreated and sEVN-treated sEVs in the samples. Data are presented as mean ± standard deviation of three isolation experiments (**p < 0.01).*

## CONCLUSIONS

In this paper, we have developed a nanofluidic device, namely the sEVN, for high-throughput sEV nanoporation, which relies on the mechanical membrane disruption to introduce transient nanopores on sEV membrane. Plasma bonding a PDMS layer with interdigital microchannels to a glass substrate with an array of nanochannels results in a rapid prototypable, leak-tight, low-cost and disposable device. Due to the layout of 30,000 equivalent nanochannels, a sample throughput of 8 μL $min^{-1}$ is demonstrated. Meanwhile, sEV concentrations before and after being treated by the sEVN is $6.4 \times 10^9$ sEVs $mL^{-1}$ and $6.1 \times 10^9$ sEVs $mL^{-1}$, respectively, corresponding to a sample recovery yield of 95.3%. Therefore, we believe that our device that dynamically exploits the nanofluidic technologies on sEV nanoporation could evolve in a tool to engineering sEVs as a universal drug delivery vehicle in biomedical research fields. Ongoing experiments on using this technique to prepare cargo-loaded sEVs are being performed, preliminary proof of concept results on intracellular delivery with the sEVN-processed sEVs verify the potentials of the technique for therapeutic applications.

## ACKNOWLEDGEMENTS

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## REFERENCES

[1] H. Shao *et al.*, "New Technologies for Analysis of Extracellular Vesicles", *Chem. Rev.*, vol. 118, pp. 1917-1950, 2018.

[2] G. Jia *et al.*, "NRP-1 Targeted and Cargo-Loaded Exosomes Facilitate Simultaneous Imaging and Therapy of Glioma in Vitro and in Vivo", *Biomaterials*, vol. 178, pp. 302-316, 2018.

[3] H. Peng *et al.*, "Exosome: A Significant Nano-Scale Drug Delivery Carrier", *J. Mater. Chem. B*, vol. 8, pp. 7591-7608, 2020.

[4] Y. Xu, "Nanofluidics: A New Arena for Materials Science", *Adv. Mater.*, vol. 30, p. 1702419, 2017.

[5] S. Park *et al.*, "A Method for Nanofluidic Device Prototyping Using Elastomeric Collapse", *Proc. Natl. Acad. Sci. U.S.A.*, vol. 106, pp. 15549-15554, 2009.

[6] M. Kim *et al.*, "Cracking-Assisted Photolithography for Mixed-Scale Patterning and Nanofluidic Applications", *Nat. Commun.*, vol. 6, p. 6247, 2015.

[7] R. Booth *et al.*, "A Multiple-Channel, Multiple-Assay Platform for Characterization of Full-Range Shear Stress Effects on Vascular Endothelial Cells", *Lab Chip*, vol. 14, pp. 1880-1890, 2014.

[8] S. R. Oh, "Thick Single-Layer Positive Photoresist Mold and Poly(dimethylsiloxane) (PDMS) Dry Etching for the Fabrication of a Glass-PDMS-Glass Microfluidic Device", *J. Micromech. Microeng.*, vol. 18, p. 115025, 2008.

[9] M. S. Kim *et al.*, "Development of Exosome-Encapsulated Paclitaxel to Overcome MDR in Cancer Cells", *Nanomedicine: NBM*, vol. 12, pp. 655-664, 2016.

[10] W. Xu *et al.*, "On-Demand in Situ Generation of Oxygen in a Nanofluidic Embedded Planar Microband Electrochemical Reactor", *Microfluid. Nanofluid.*, vol. 19, pp. 1181-1189, 2015.

## CONTACT

*H. Yang, tel: +86-0755-86392675; hui.yang@siat.ac.cn



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